

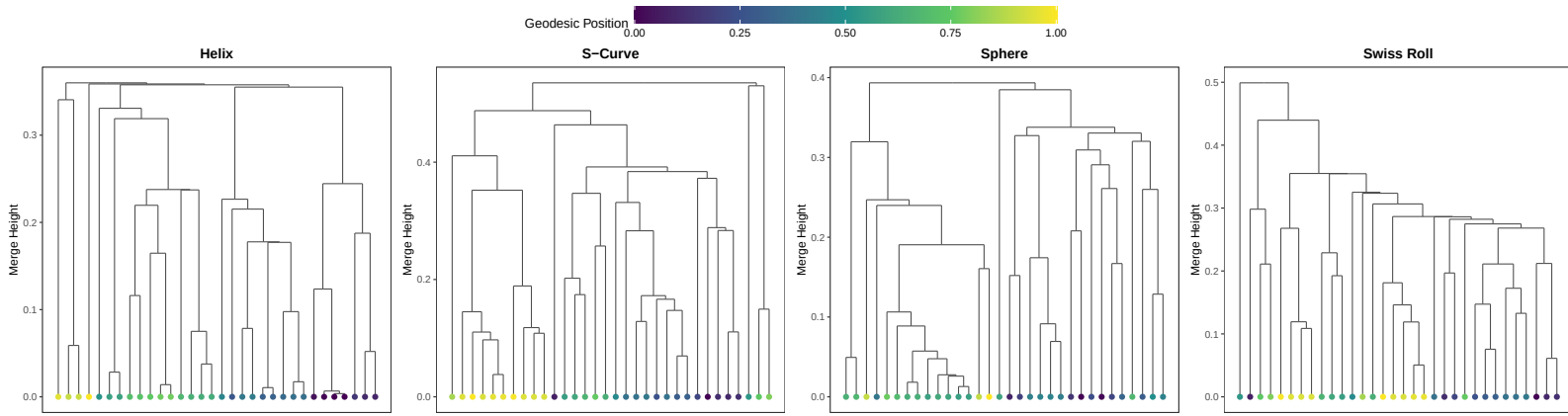


Figure 1: Single-trial AHC dendrograms for four artificial datasets. We sample $N = 300$ points from each manifold (Helix, S-Curve, Sphere, Swiss Roll) and build a single AHC tree via random subsampling with subsample size $\psi = 32$ and single linkage. Each leaf represents one of the 32 randomly subsampled data points. Leaf nodes are coloured by their *geodesic position*: arc length for Helix; first intrinsic coordinate for S-Curve and Swiss Roll; azimuthal angle for Sphere, normalised to $[0, 1]$ and mapped to a viridis colour scale. *Analysis:* For the Helix and S-Curve, the dendrograms show that geodesically nearby points (similar colours) are merged at low heights, while geodesically distant points (contrasting colours) are only joined at higher levels. The Sphere and Swiss Roll show broader colour mixing at higher merge levels, but local geodesic clusters are still clearly visible within subtrees, confirming that the AHC tree preserves geodesic locality even in a single trial. This locality preservation is the fundamental structural property that enables AGD and AGD-SS to approximate geodesic distance: because geodesically nearby points are co-assigned to the same or nearby subtrees, the tree-based distance (merge height or subtree summation) provides a meaningful proxy for geodesic separation.

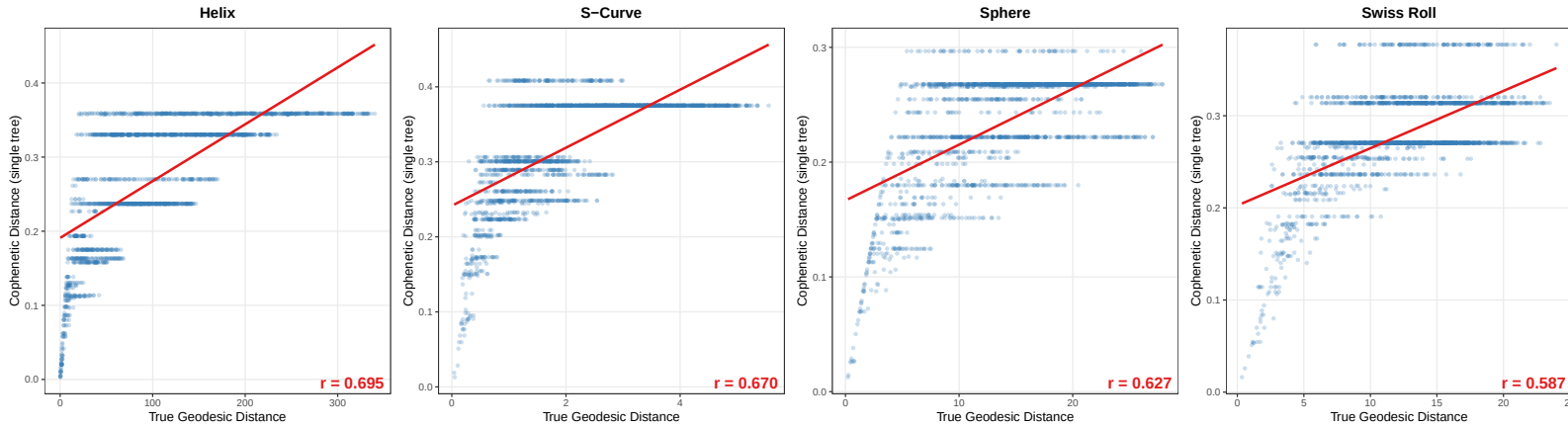


Figure 2: Cophenetic distance versus true geodesic distance (single AHC tree). For each manifold ($N = 300$), a single AHC tree is constructed with $\psi = 64$ and single linkage. For every pair of subsampled points ($\binom{64}{2} = 2016$ pairs), we compute the **cophenetic distance (the merge height at which the two points first join in the dendrogram)** and the true geodesic distance. Each dot is one pair; the red line is a linear regression fit, with Pearson r annotated. *Note:* Pearson r measures *linear* association only. A moderate r does not imply failure to preserve the geodesic *rank ordering*. We report Pearson r , providing a conservative guarantee. *Analysis:* All four manifolds exhibit clear positive correlations: Helix $r = 0.695$, S-Curve 0.670 , Sphere 0.627 , Swiss Roll 0.587 . The correlation ranking (Helix > S-Curve > Sphere > Swiss Roll) reflects increasing geometric complexity: the Helix is a simple 1-D curve with no folding; the S-Curve and Sphere introduce non-linear bend; and the Swiss Roll’s tight spiral creates regions where Euclidean-close points are geodesically distant. The scatter plots also reveal a characteristic “fan” shape: tight at small geodesic distances but widening at large distances, indicating that single-tree cophenetic estimates are more reliable for local neighbours than for distant pairs. This increasing variance at large distances motivates the ensemble averaging strategy in Figure 4.

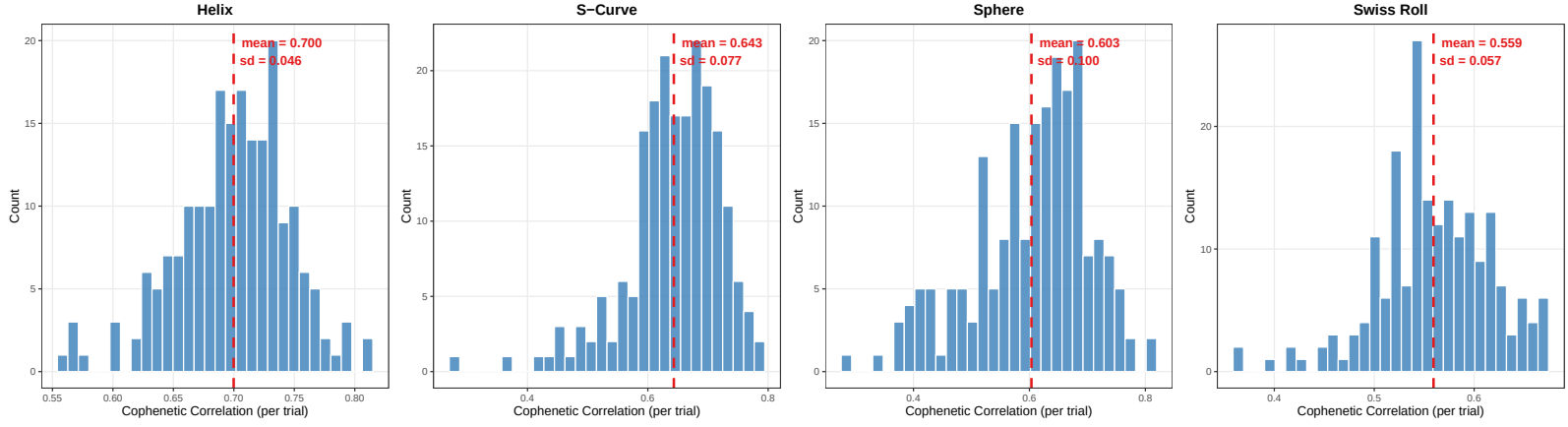


Figure 3: Distribution of cophenetic correlations across $T = 200$ Monte-Carlo trials. We repeat the single-tree cophenetic correlation measurement (as in Figure 2) independently $T = 200$ times, each with a random subsample ($\psi = 64$, single linkage). The histogram shows the distribution of Pearson r values across trials; the dashed red line marks the mean. *Analysis:* All four manifolds yield consistently positive correlations, with means ranging from 0.559 ± 0.057 (Swiss Roll) to 0.700 ± 0.046 (Helix). The Helix (mean $r = 0.700$) and S-Curve ($r = 0.643$) achieve higher correlations than the Sphere ($r = 0.604$) and Swiss Roll ($r = 0.559$), reflecting the greater geometric complexity of the latter. Crucially, the distributions are concentrated well above zero and roughly unimodal, confirming that the positive correlation is not an artefact of a single lucky draw but a *systematic* property of the AHC construction. The moderate standard deviations ($\sigma \approx 0.05$ – 0.10) reveal meaningful tree-to-tree variation, which is exactly the property that makes the ensemble-averaging strategy effective: while individual trees are noisy estimators, their *average* converges to a stable geodesic distance estimator (see Figure 4). The Sphere shows the widest spread ($\sigma = 0.100$), suggesting that random subsampling on a 2-D surface introduces more variability than on 1-D curves, yet even its worst-case trials remain positive.

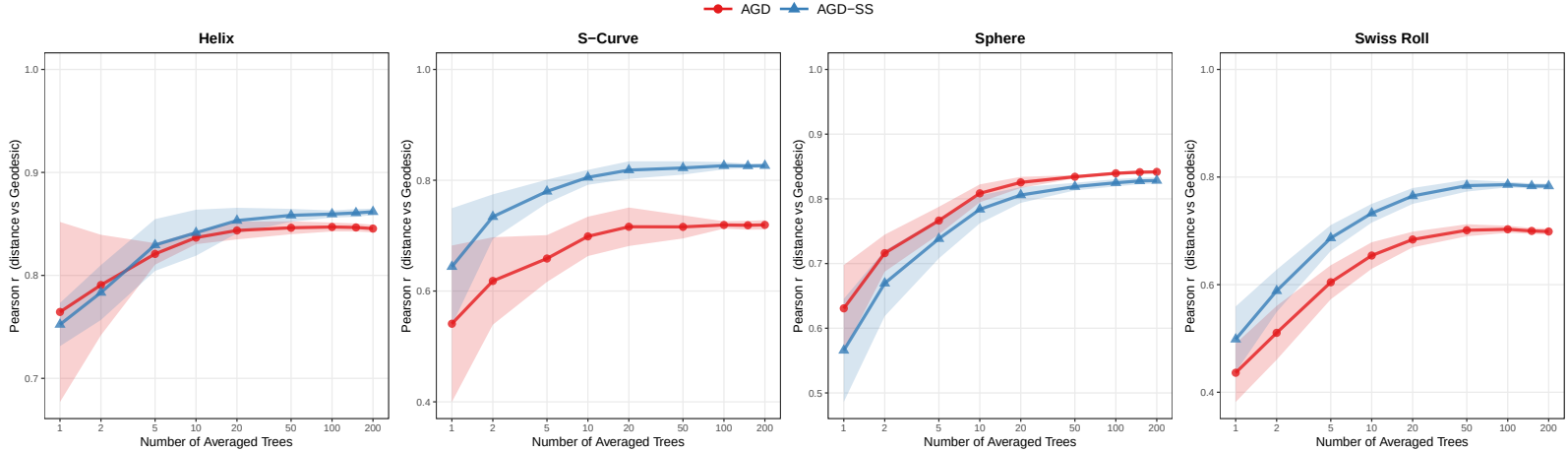


Figure 4: Aggregation effect: Pearson correlation versus number of averaged trees t . For each manifold ($N = 300$), we build forests of up to $T = 200$ independent AHC trees ($\psi = 64$, single linkage). Two distance definitions are compared: **AGD** and **AGD-SS**. Solid lines show the mean Pearson r across 10 independent repetitions; shaded bands show standard deviation, quantifying the variance due to random subsampling. *Analysis:* Both methods improve monotonically, with the steepest gains in the first 5–10 trees, after which the curves plateau, demonstrating that the ensemble-averaging mechanism is the key of quality improvement. The narrowing $\pm\sigma$ bands confirm that variance decreases as more trees are averaged, meaning the AGD estimate becomes increasingly *reproducible* with larger forests. At $t = 200$, **AGD-SS consistently outperforms AGD**: S-Curve (0.82 vs 0.72), Swiss Roll (0.77 vs 0.68), and Helix (0.86 vs 0.85), with the exception of Sphere where both reach $r \approx 0.83$. The AGD-SS advantage is most pronounced on the S-Curve and Swiss Roll, where the subtree-sum formulation starts with higher single-tree correlation and maintains the gap throughout. This suggests that counting descendant nodes (AGD-SS) extracts richer geodesic structure from each tree than summing edge weights to the LCA (AGD), particularly on manifolds with complex non-linear geometry.

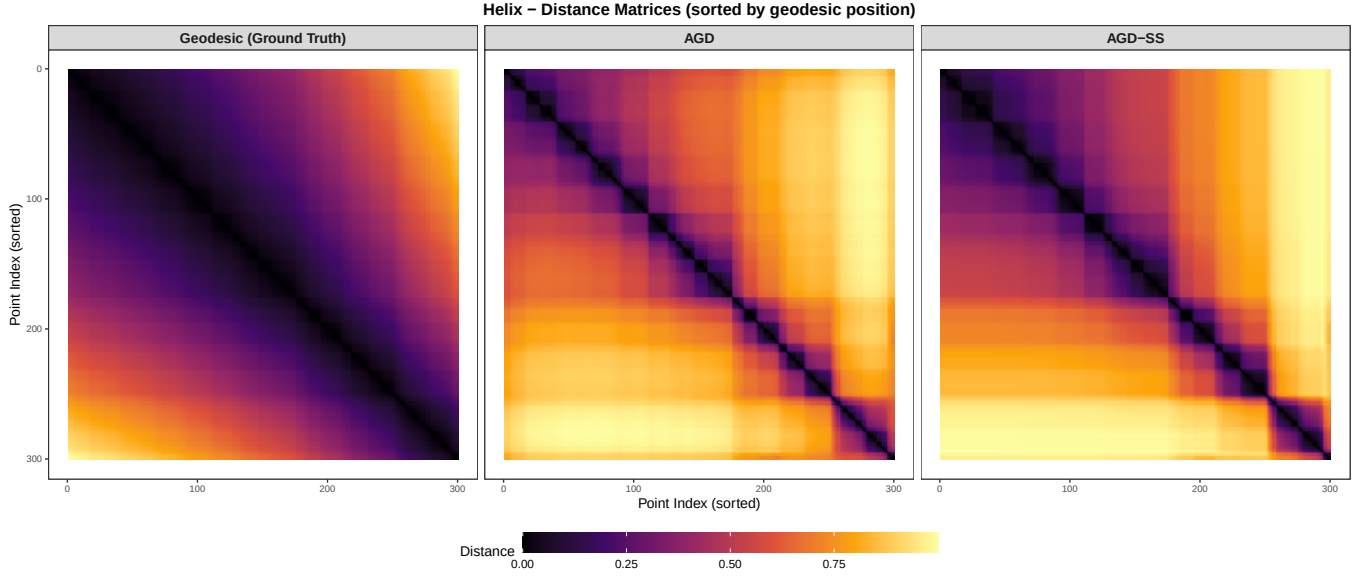


Figure 5: Distance matrix heatmaps for the Helix manifold. The full 300×300 pairwise distance matrices are computed using $\psi=64$ and $t=200$ trees with single linkage. Rows and columns are sorted by geodesic position (arc length), and each matrix is normalised to $[0, 1]$. Left: ground-truth geodesic distance; Centre: AGD ($r = 0.848$); Right: AGD-SS ($r = 0.863$). All three heatmaps exhibit a clear diagonal-band structure, where nearby points have small distances and distant points have large distances. Both AGD and AGD-SS closely reproduce the geodesic distance pattern; AGD-SS achieves slightly higher correlation, consistent with the aggregation analysis in Figure 4.

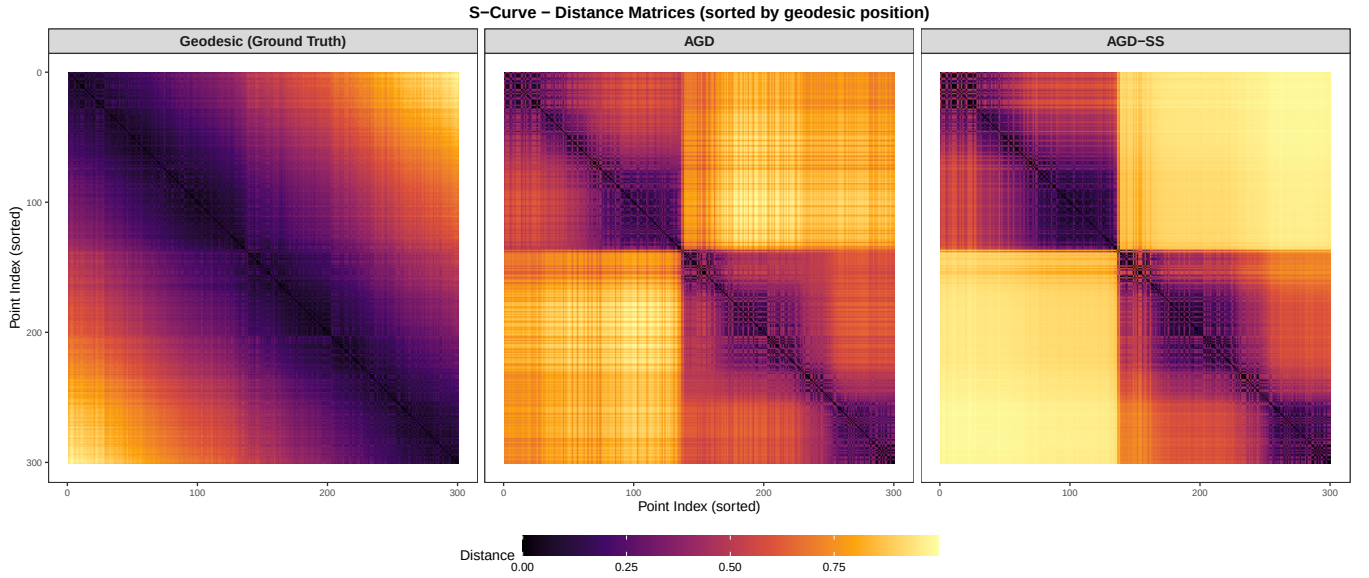


Figure 6: Distance matrix heatmaps for the S-Curve manifold. Same protocol as Figure 5: $N=300$, $\psi=64$, $t=200$ trees, single linkage. The geodesic distance is computed via k -NN graph shortest paths ($k=10$). Matrices are sorted by the first intrinsic coordinate. AGD ($r=0.716$) reproduces the broad structure but shows noticeable blurring at intermediate distances. AGD-SS ($r=0.821$) provides a substantially sharper reconstruction with a $+0.105$ improvement, demonstrating that the subtree-sum formulation captures finer geodesic detail on manifolds with non-linear geometry.

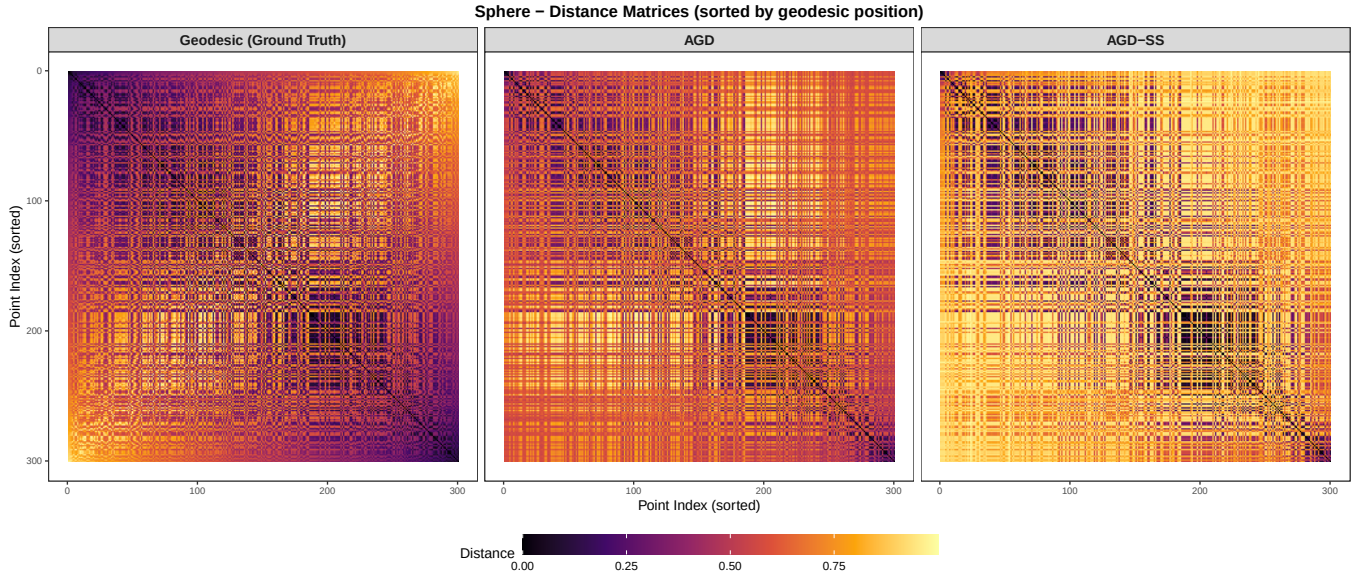


Figure 7: Distance matrix heatmaps for the Sphere manifold. Same protocol: $N = 300$, $\psi = 64$, $t = 200$ trees, single linkage. Matrices are sorted by azimuthal angle. The Sphere’s heatmap displays a smooth gradient from the diagonal outward. Both AGD ($r = 0.838$) and AGD-SS ($r = 0.827$) faithfully reproduce this pattern. This is the only manifold where AGD slightly outperforms AGD-SS, likely because the Sphere’s uniform curvature and simple distance structure are well-suited to the path-to-LCA formulation.

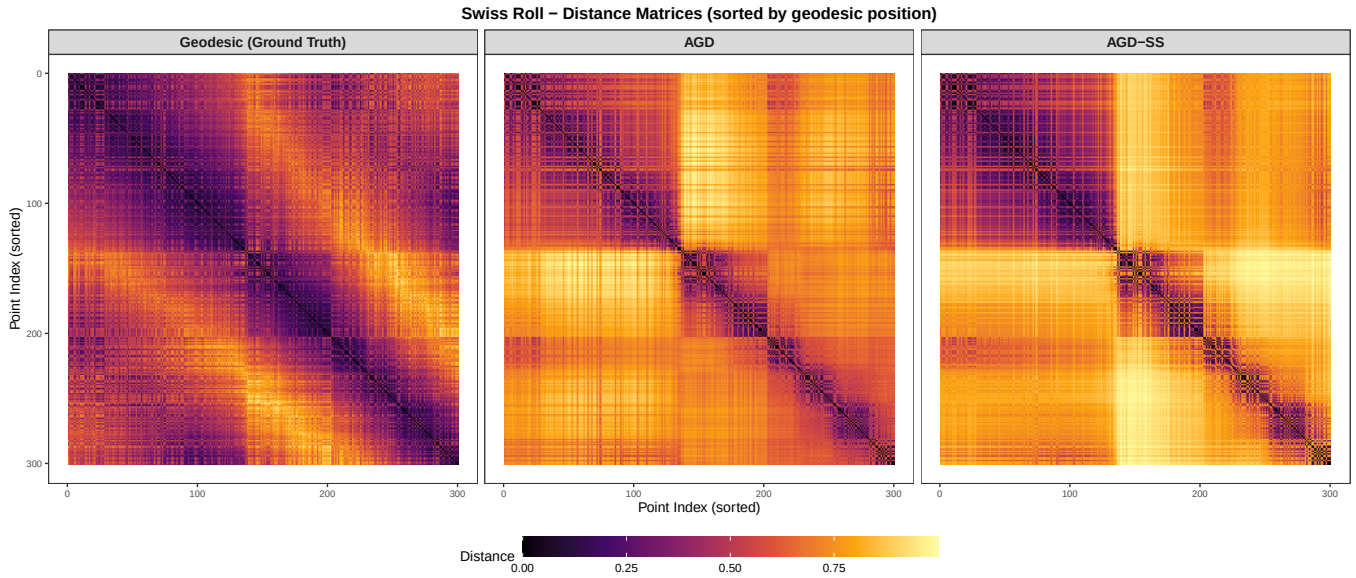


Figure 8: Distance matrix heatmaps for the Swiss Roll manifold. Same protocol: $N = 300$, $\psi = 64$, $t = 200$ trees, single linkage. The geodesic distance is computed via k -NN graph shortest paths ($k = 10$); matrices are sorted by the unrolled position. The Swiss Roll is the most challenging manifold due to its tight spiral, where Euclidean-close points from different layers are geodesically distant. AGD ($r = 0.676$) captures the overall trend but shows substantial noise. AGD-SS ($r = 0.767$, $+0.091$) is markedly more robust to this “short-circuit” geometry, achieving the largest improvement over AGD among all four manifolds.

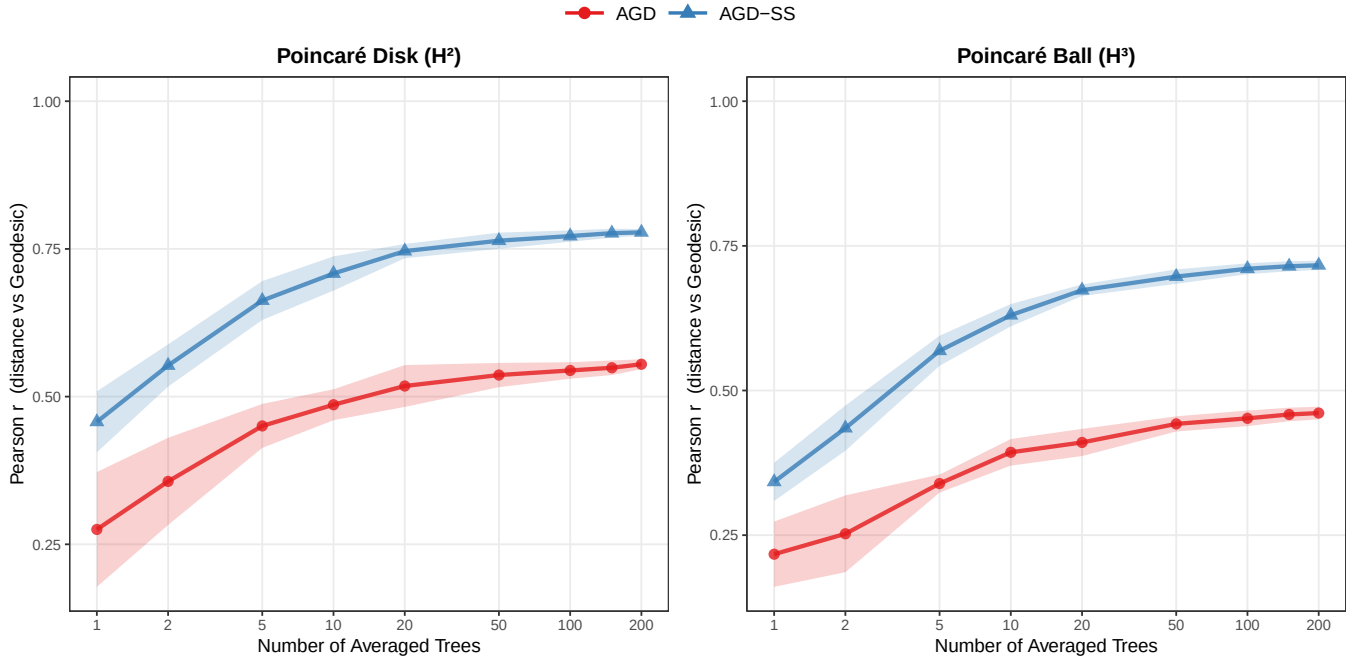


Figure 9: Aggregation effect on hyperbolic manifolds ($\psi = 64$, single linkage, 10 repetitions, $\pm\sigma$ bands). Red: **AGD**; Blue: **AGD-SS**. *Analysis:* On H^2 , AGD-SS dramatically outperforms AGD: $r = 0.78$ vs. 0.56 at $t = 200$, a $+0.22$ gap. This is the largest AGD-SS advantage among all manifolds tested. The tree-like geometry of hyperbolic space (where subtree sizes encode the exponential branching structure) is particularly well-suited to the subtree-sum formulation. On H^3 , aggregation similarly improves both methods: AGD-SS reaches $r = 0.72$, AGD $r = 0.46$ at $t = 200$, maintaining a large $+0.26$ gap. The consistent AGD-SS superiority in both 2-D and 3-D hyperbolic space demonstrates that the subtree-sum formulation is robustly better suited to tree-like (negatively curved) geometries, regardless of dimension. The narrowing bands confirm that estimator variance decreases with more trees.

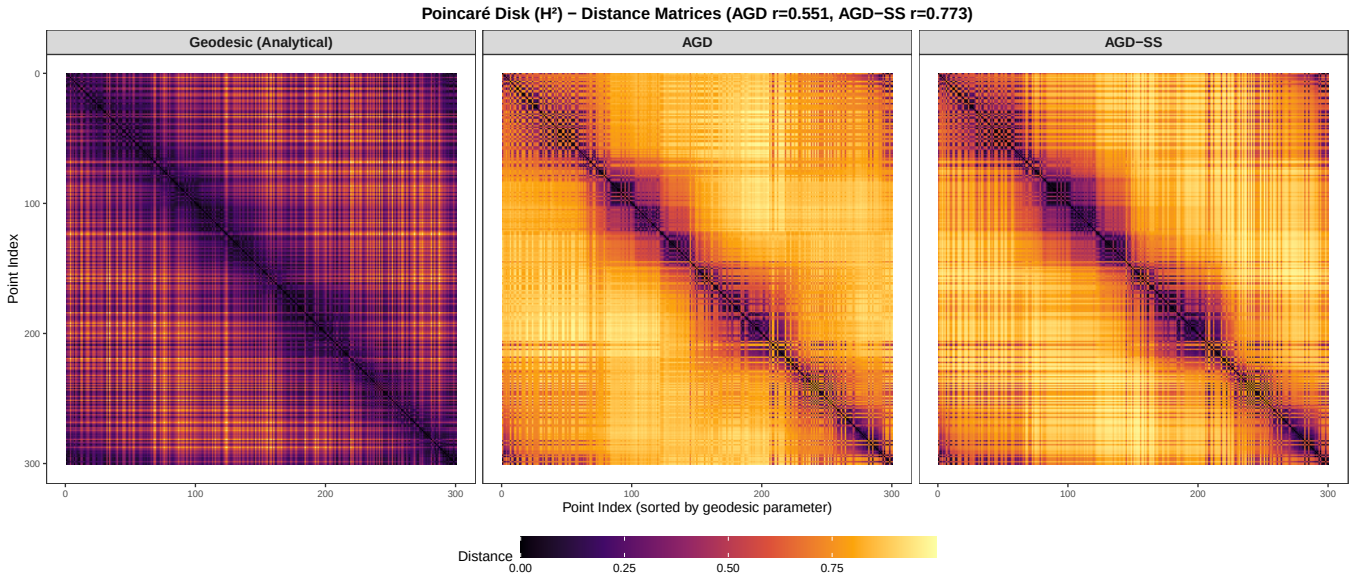


Figure 10: Distance matrix heatmaps for Poincaré Disk (H^2). 300×300 pairwise distance matrices ($\psi = 64$, $t = 200$, single linkage), sorted by angular position and normalised to $[0, 1]$. Left: **exact hyperbolic geodesic**; Centre: **AGD** ($r = 0.551$); Right: **AGD-SS** ($r = 0.773$). *Analysis:* The geodesic heatmap reveals the smooth angular structure of the Poincaré disk. AGD shows noticeable spreading, reflecting the difficulty of approximating the exponentially-scaled hyperbolic metric from Euclidean-space partitions. AGD-SS reconstructs the diagonal-band pattern far more faithfully, achieving a $+0.222$ improvement over AGD.

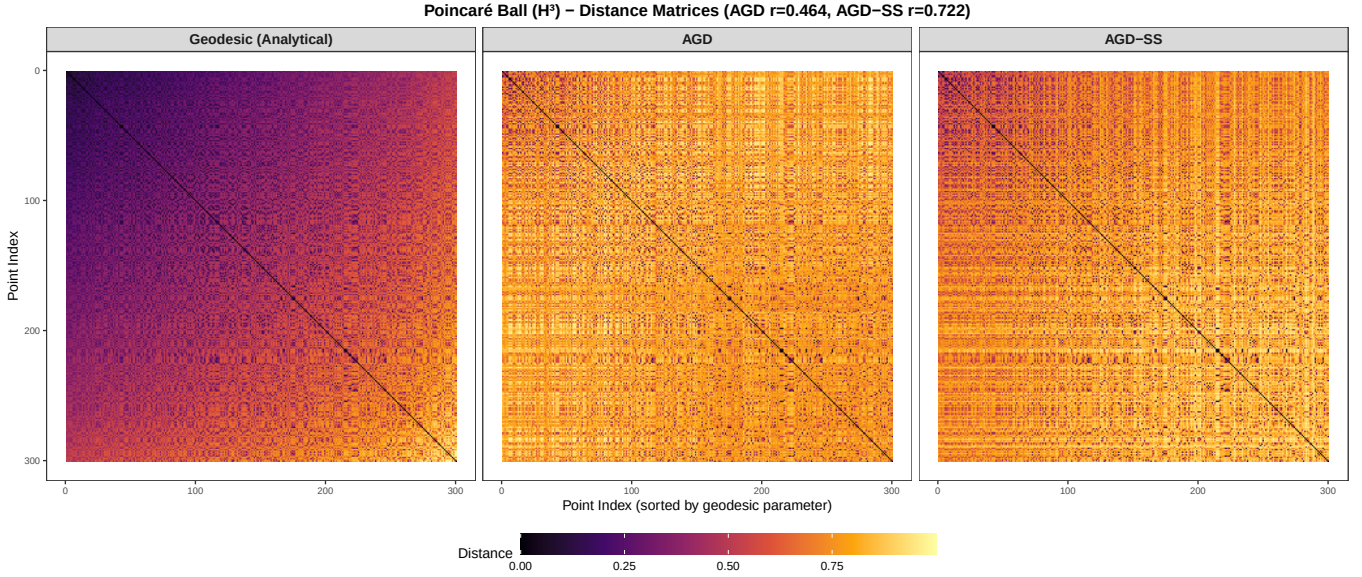


Figure 11: Distance matrix heatmaps for Poincaré Ball (H^3). 300×300 pairwise distance matrices ($\psi=64$, $t=200$, single linkage), sorted by radial distance from the origin and normalised to $[0, 1]$. Left: **exact hyperbolic geodesic**; Centre: **AGD** ($r=0.464$); Right: **AGD-SS** ($r=0.722$). *Analysis:* Moving from H^2 to H^3 reduces absolute correlations (AGD: $0.551 \rightarrow 0.464$; AGD-SS: $0.773 \rightarrow 0.722$), reflecting the increased difficulty of partitioning a 3-D volume vs. a 2-D disk. Crucially, AGD-SS maintains a large $+0.258$ advantage over AGD, even larger than on H^2 . Both AGD and AGD-SS capture the broad diagonal-band structure, but AGD-SS recovers finer off-diagonal patterns more faithfully. The consistent superiority of AGD-SS across both H^2 and H^3 confirms that the subtree-sum formulation is robustly better suited to the tree-like (negatively curved) geometry of hyperbolic space.

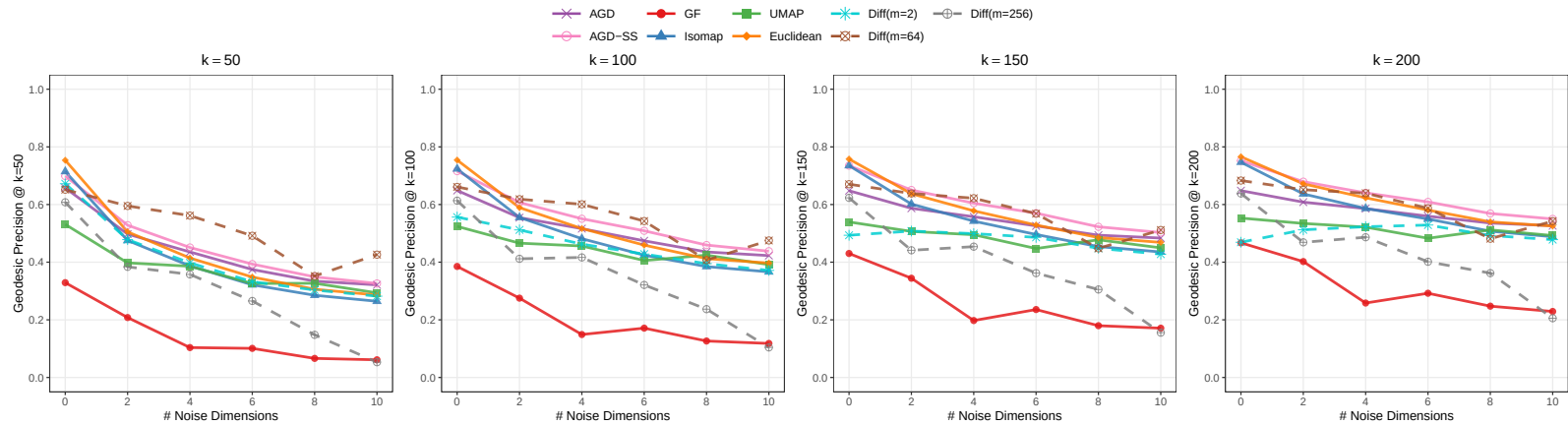


Figure 12: Geodesic Precision on the Poincaré Ball (H^3) at $k=50, 100, 150, 200$ ($N=1000$, noise constant = 20). Here we use the same parameter setting as in the main paper (Figure 3). Each panel shows a different evaluation neighbourhood size k . *Analysis:* At **zero noise**, Euclidean (0.75) and Isomap (0.71) achieve the highest precision at $k=50$, but both degrade sharply with noise (to 0.29 and 0.26 at 10 noise dims). **Diffusion** ($m=64$) is the most noise-robust method when $k \leq 100$, but its performance collapses at higher noise levels and larger k values. **AGD-SS** performs better when k is larger: $0.70 \rightarrow 0.33$ at $k=150$ and $0.75 \rightarrow 0.55$ at $k=200$, consistently outperforming AGD at every k and noise level. Diffusion with $m=64$ is the best when $k \leq 100$. GF performs poorly throughout (≤ 0.47), confirming its difficulty with hyperbolic geometry.

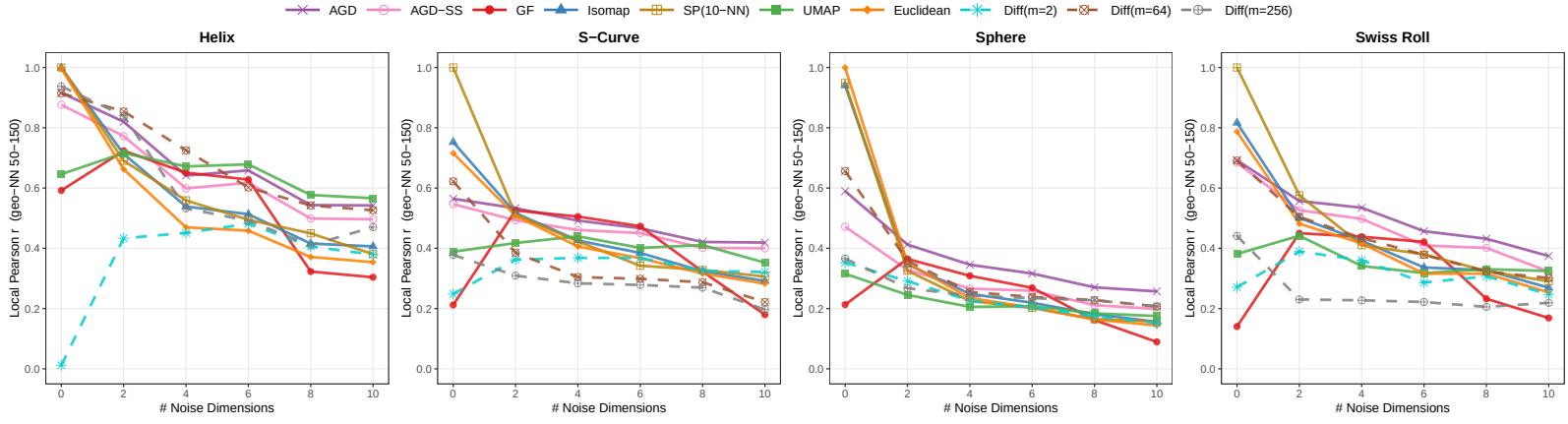


Figure 13: Pearson correlation with true geodesic distance on four manifolds ($N = 1000$, noise constant = 20, noise dims $\in \{0, 2, 4, 6, 8, 10\}$). In order to align with the same setting of Figure 3 in our main paper, for each point x_i we identify its 50th–150th nearest neighbours under the true geodesic (a set S_i of 101 points) and compute the Pearson r between the estimated and true geodesic distances to S_i . The plotted value is the average over all N points. Ten methods are compared: the nine from Figure 3 in the main paper, plus **SP(10-NN)**, which computes shortest-path distances on a 10-NN graph without the classical MDS embedding step used by Isomap. *Analysis:* **Helix:** AGD ($0.92 \rightarrow 0.54$) and Diff-64 ($0.91 \rightarrow 0.53$) are the most noise-robust, followed closely by UMAP ($0.65 \rightarrow 0.57$). Isomap, SP(10-NN), and Euclidean start high (~ 1.0) but degrade sharply. UMAP is the best method when noise dims ≥ 6 . **S-Curve:** AGD and AGD-SS are the most robust methods; **Sphere:** Euclidean, Isomap, and SP(10-NN) achieve near-perfect local r at zero noise, but collapse with noise. AGD is the most stable ($0.59 \rightarrow 0.26$), followed by AGD-SS. **Swiss Roll:** AGD outperforms every other method at all noise levels. AGD-SS is a close second. SP(10-NN) tracks Isomap closely throughout, confirming that the CMDS step has negligible impact on local distance quality. In general, AGD and AGD-SS show superior noise robustness across all four manifolds, with AGD particularly strong on the Swiss Roll, while Euclidean, Isomap, and SP(10-NN) are highly sensitive to noise, and Diffusion’s performance varies with the choice of m and the manifold.

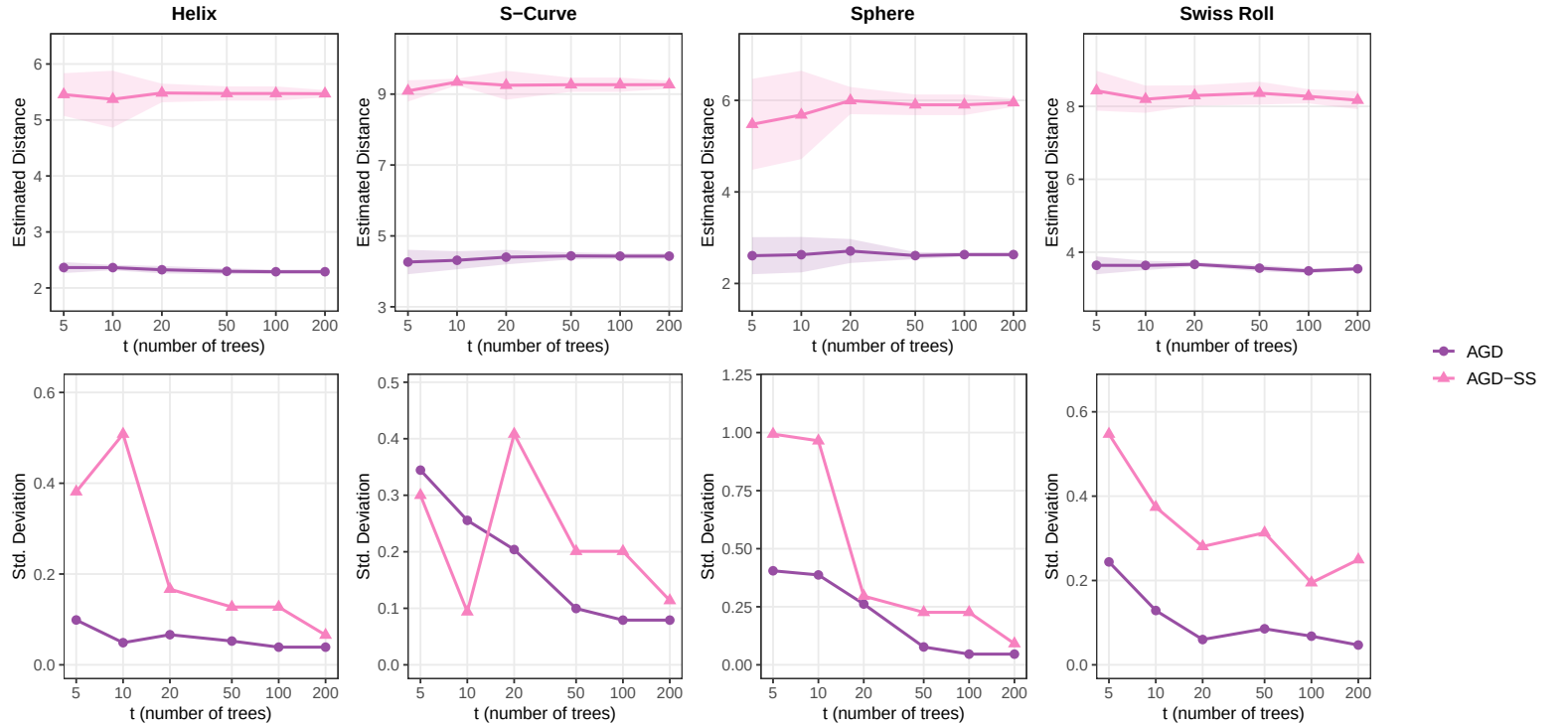


Figure 14: Pointwise concentration of AGD and AGD-SS for a single pair of points (5 independent runs). One pair (i, j) was randomly selected and held fixed across all runs. **Top row:** mean estimated pairwise distance $\pm$ one standard deviation (shaded). **Bottom row:** standard deviation of the estimate across the 5 runs. *Analysis:* On every manifold, both AGD and AGD-SS concentrate rapidly: the standard deviation drops sharply from $t = 5$ to $t = 20$ – 50 and continues to decrease towards $t = 200$. Note that AGD and AGD-SS produce distance values on different scales (AGD-SS is generally larger) because they use different aggregation strategies, yet both exhibit the same monotone variance reduction, confirming that individual pairwise distances concentrate with increasing t .

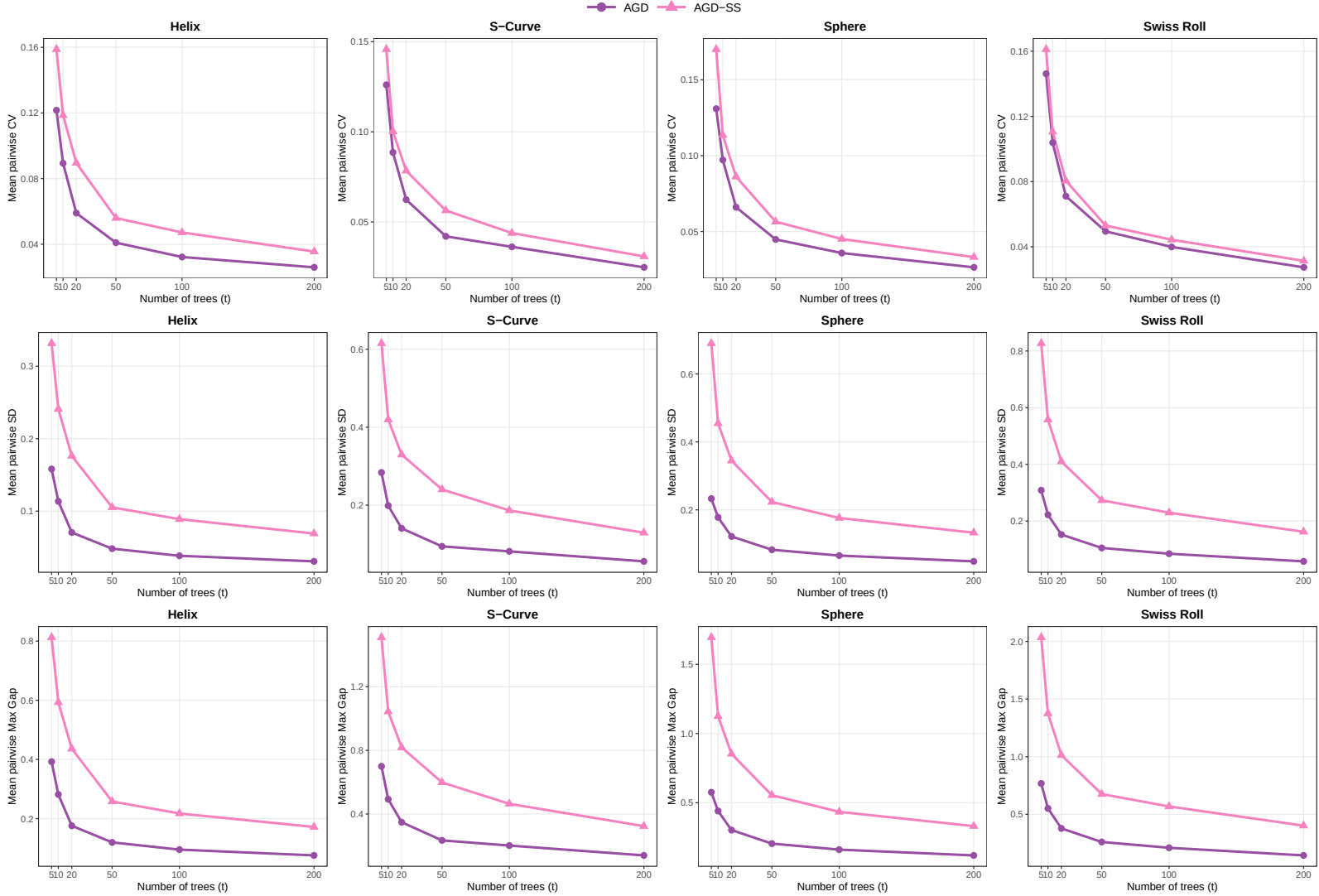


Figure 15: Pairwise concentration across all $\binom{N}{2}$ point pairs ($N = 300$, $\psi = 64$, single linkage, $R = 5$ independent runs). Let $\hat{d}_{ij}^{(r)}$ denote the estimated distance between points i and j in the r -th independent run ($r = 1, \dots, R$). Define the per-pair mean $\mu_{ij} = \frac{1}{R} \sum_{r=1}^R \hat{d}_{ij}^{(r)}$, standard deviation $\sigma_{ij} = [\frac{1}{R-1} \sum_{r=1}^R (\hat{d}_{ij}^{(r)} - \mu_{ij})^2]^{1/2}$, coefficient of variation $CV_{ij} = \sigma_{ij} / \mu_{ij}$, and max gap $MG_{ij} = \max_r \hat{d}_{ij}^{(r)} - \min_r \hat{d}_{ij}^{(r)}$. All statistics below are averaged over the $\binom{N}{2}$ pairs. **Top row (Mean Pairwise CV):** $\overline{CV}(t) = \binom{N}{2}^{-1} \sum_{i < j} CV_{ij}$, the mean coefficient of variation as a function of the number of trees t . **Middle row (Mean Pairwise SD):** $\bar{\sigma}(t) = \binom{N}{2}^{-1} \sum_{i < j} \sigma_{ij}$, the mean raw standard deviation (unnormalised) across all pairs. **Bottom row (Mean Pairwise Max Gap):** $\overline{MG}(t) = \binom{N}{2}^{-1} \sum_{i < j} MG_{ij}$, the mean of the per-pair range (maximum minus minimum) over R runs, measuring the worst-case single-run deviation for a typical pair. *Analysis:* All three metrics decrease monotonically with t for both AGD and AGD-SS across all four manifolds, confirming concentration of the distance estimator. The mean CV drops from ~ 12 – 16% at $t = 5$ to ~ 2.5 – 3.5% at $t = 200$. AGD consistently attains lower CV, SD, and max gap than AGD-SS (the subtree-sum formulation has larger dynamic range), but both estimators concentrate at comparable rates.

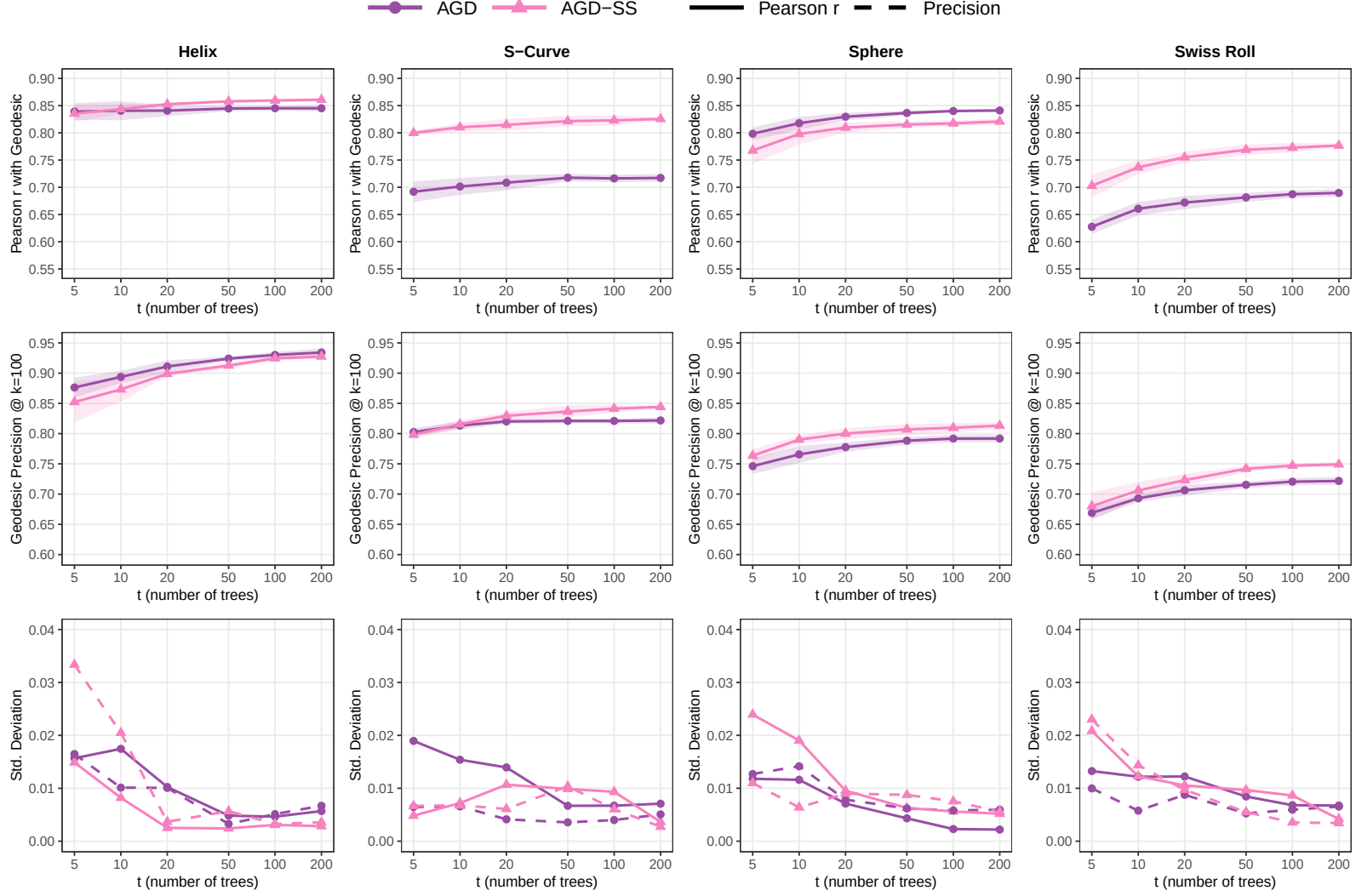


Figure 16: Concentration of AGD and AGD-SS as a function of t (number of trees) (5 independent runs). **Top row:** Pearson r with the true geodesic distance matrix. **Middle row:** Geodesic Precision at $k = 100$. **Bottom row:** Standard deviation across the 5 runs (solid = Pearson r , dashed = Precision). Shaded bands in the top two rows denote standard deviation. *Analysis:* Both metrics converge rapidly: most of the improvement is attained by $t = 20$ – 50 trees, with marginal gains beyond that. Standard deviations shrink monotonically with t , falling below 0.01 on every dataset by $t = 50$, confirming that both estimators are stable and reproducible. AGD-SS outperforms AGD in Pearson r on all four manifolds except the Sphere (where AGD is slightly higher in r). In geodesic precision, AGD-SS is consistently better on S-Curve, Sphere, and Swiss Roll; on the Helix, AGD achieves slightly higher precision but AGD-SS has a higher Pearson r .